

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 1: Introduction

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September 15, 2026

1 Keywords

- Course organisation, vectors, matrices, transposes, determinants, gradients, random variables, expectations

2 Main Points

2.1 About the Course

- The course is dual coded as AICE3001 (third year Artificial Intelligence) and AICE6002 (MSc Machine Learning), with slightly different assessment for the two
- Notes are on Moodle and at <https://ecs-vlc.github.io/aice3001/>
- **Why learn mathematics when LLMs can do so much?** Because you need to be able to trust your own judgement. LLMs are useful, but if you cannot check what they tell you, you are disempowered
- Learning machines are notoriously hard to understand. Mathematics lets us reason rigorously, build intuition we can rely on, describe the world precisely, and gives us a set of tools that turn out to be hugely useful
- Mathematics is a language that takes time to learn. Everyone can crack the code, but only by working with it—which is why the tutorials should be treated as compulsory
- Students start from very different places. If you are strong, become stronger by helping others; if you are starting from a weak base, judge yourself by your progress rather than by comparison
- **The golden rule of mathematics: understand every step.** When you don't understand a step, stop, break the problem down, and ask. Asking "stupid" questions is how everyone learns
- The course covers calculus, vector spaces, convexity, optimisation, probability, statistics and information theory: the foundations of most of machine learning

2.2 What We Assume You Know

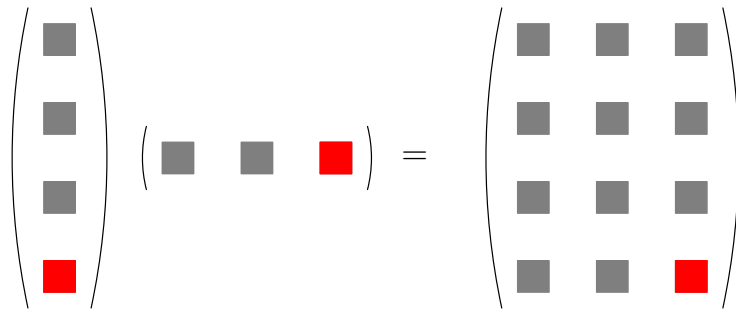
- A-level mathematics and some university mathematics: functions such as $f(x) = x^2 + 3$, simple algebra, and elementary calculus such as $f'(x) = 2x$
- The rest of this note summarises the assumed background. We will revisit most of it in more depth, but if it is unfamiliar you should revise it now, because other machine learning modules rely on it

2.3 Vectors

- Vectors in $\mathbb{R}^n$ are written in bold and are *column* vectors,

$$\mathbf{x} = \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix}$$

- The corresponding row vector is the transpose, $\mathbf{x}^\top = (1.2, 2.3, 4.7)$. Keeping track of rows and columns lets us write vector equations succinctly and unambiguously
- The *inner product* (dot product) of two vectors of the same length is $\mathbf{a}^\top \mathbf{b} = \sum_i a_i b_i$. It encodes the angle between them: $\mathbf{a}^\top \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$
- The *outer product* $\mathbf{a} \mathbf{b}^\top$ is a matrix with elements $a_i b_j$, defined for vectors of any sizes



- The *Euclidean norm* (2-norm) measures size,

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{\mathbf{x}^\top \mathbf{x}}$$

Other norms exist; this one is intimately tied to the inner product

2.4 Matrices

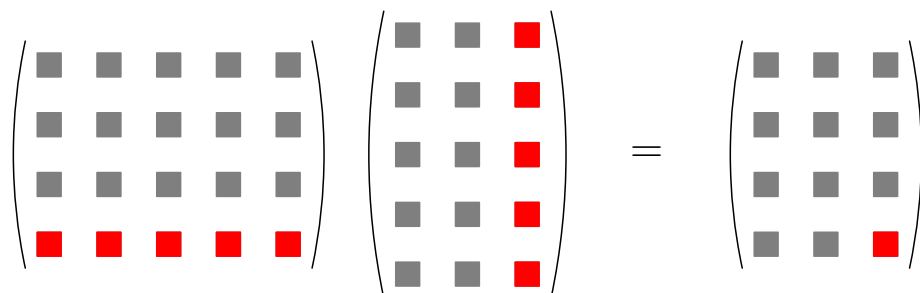
- An $m \times n$ matrix, $\mathbf{M} \in \mathbb{R}^{m \times n}$, has m rows and n columns, and maps vectors in $\mathbb{R}^n$ to vectors in $\mathbb{R}^m$:

$$\mathbf{M} \mathbf{x} = \begin{pmatrix} 1.1 & -2.1 & 6.2 \\ 9.3 & 3.1 & -4.4 \end{pmatrix} \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix} = \begin{pmatrix} 25.63 \\ -2.39 \end{pmatrix}$$

- Each output element is a linear sum of the inputs, $b_i = \sum_j M_{ij} a_j$, so for any real c

$$\begin{aligned} \mathbf{M}(c \mathbf{a}) &= c(\mathbf{M} \mathbf{a}) \\ \mathbf{M}(\mathbf{a} + \mathbf{b}) &= \mathbf{M} \mathbf{a} + \mathbf{M} \mathbf{b} \end{aligned}$$

- If $\mathbf{A} \in \mathbb{R}^{\ell \times m}$ and $\mathbf{B} \in \mathbb{R}^{m \times n}$ then $\mathbf{A} \mathbf{B}$ is the mapping "first apply $\mathbf{B}$, then $\mathbf{A}$ ", $\mathbf{A} \mathbf{B} \mathbf{x} = \mathbf{A}(\mathbf{B} \mathbf{x})$. It is an $\ell \times n$ matrix computed by matrix multiplication (rows times columns)



- Matrix multiplication is associative, $(\mathbf{A}\mathbf{B})\mathbf{C} = \mathbf{A}(\mathbf{B}\mathbf{C})$, but in general not commutative, $\mathbf{A}\mathbf{B} \neq \mathbf{B}\mathbf{A}$
- A column vector can be treated as an $n \times 1$ matrix, so matrix-vector multiplication is a special case of matrix multiplication

2.5 Transposes

- The transpose swaps rows and columns, $[\mathbf{M}^\top]_{ij} = [\mathbf{M}]_{ji}$, and $(\mathbf{M}^\top)^\top = \mathbf{M}$
- A little book keeping shows $(\mathbf{A}\mathbf{B})^\top = \mathbf{B}^\top \mathbf{A}^\top$
- A matrix is *symmetric* if $\mathbf{M} = \mathbf{M}^\top$. A 1×1 matrix is always symmetric, which gives two identities we will use constantly:

$$\mathbf{a}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{a}$$

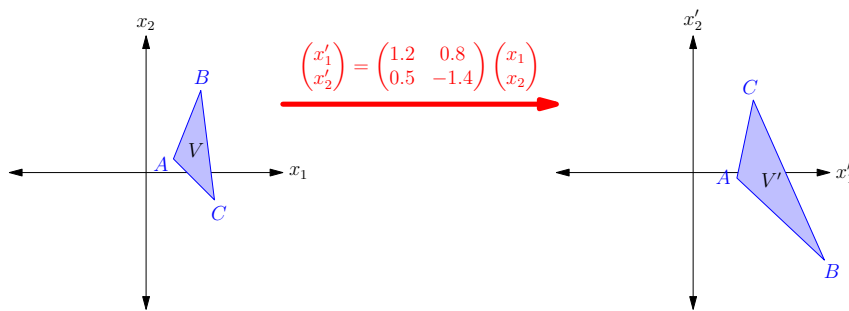
$$\mathbf{a}^\top \mathbf{M} \mathbf{b} = \mathbf{b}^\top \mathbf{M}^\top \mathbf{a}$$

2.6 Determinants

- Treat each point of a region of volume V in $\mathbb{R}^n$ as a vector and map it by a square matrix $\mathbf{M}$. The image has volume V' , and

$$\det(\mathbf{M}) = \pm \frac{V'}{V}$$

where the sign records whether the mapping includes a reflection



- Because $\mathbf{A}\mathbf{B}$ means "apply $\mathbf{B}$ then $\mathbf{A}$ ", volume scalings multiply: $\det(\mathbf{A}\mathbf{B}) = \det(\mathbf{A}) \det(\mathbf{B})$
- Determinants are only defined for square matrices; a non-square matrix maps between spaces of different dimension, where volumes cannot be compared
- A *singular* matrix squashes space onto a lower dimensional subspace, so its determinant is 0

2.7 Multivariate Functions and Gradients

- Machine learning uses functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ heavily, and very often scalar functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$, such as a learning machine $\hat{f}(\mathbf{x}, \mathbf{w})$ or a loss function $L(\mathbf{w})$
- A function is *linear* if $f(c\mathbf{x}) = cf(\mathbf{x})$ and $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$
- The *gradient* of a scalar function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the column vector of partial derivatives

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix}$$

- It is orthogonal to the contours of f and points in the direction in which f increases fastest

2.8 Probability

- A *random variable* assigns a number to the outcome of a random event: 0 for tails and 1 for heads is a discrete random variable; the temperature in Hyde Park at noon tomorrow is a continuous one
- For a discrete random variable the *probability mass function* $\mathbb{P}[X = x]$ lies between 0 and 1 and $\sum_{x \in \mathcal{X}} \mathbb{P}[X = x] = 1$
- For a continuous random variable the probability of any single value is zero, and we use the *probability density*

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}[x < X \leq x + \delta x]}{\delta x}$$

- The *expectation* of a function $g(X)$ is its average value,

$$\mathbb{E}[g(X)] = \sum_{x \in \mathcal{X}} \mathbb{P}[X = x] g(x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int f_X(x) g(x) dx$$

- Expectation is linear, $\mathbb{E}[c g(X)] = c \mathbb{E}[g(X)]$ and $\mathbb{E}[g(X) + h(X)] = \mathbb{E}[g(X)] + \mathbb{E}[h(X)]$, and the expectation of a constant is itself, $\mathbb{E}[c] = c$

3 Exercises

These are warm-ups to check the assumed background.

3.1 Matrix multiplication does not commute

- Compute $\mathbf{A}\mathbf{B}$ and $\mathbf{B}\mathbf{A}$ for

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$$

3.2 Transpose of a product

- Prove $(\mathbf{A}\mathbf{B})^\top = \mathbf{B}^\top \mathbf{A}^\top$ by writing out the elements

3.3 Determinants as areas

- Compute the determinant of $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ and say what it tells you about the image of the unit square

3.4 Expectations

- For a fair six-sided die X , compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$ and the variance $\mathbb{E}[X^2] - \mathbb{E}[X]^2$
- Use the definition of expectation to show $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$

4 Answers

4.1 Matrix multiplication does not commute

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$$\mathbf{A}\mathbf{B} = \begin{pmatrix} 7 & 2 \\ 3 & 1 \end{pmatrix} \quad \mathbf{B}\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}$$

which are different

4.2 Transpose of a product

- $[(\mathbf{A}\mathbf{B})^T]_{ij} = [\mathbf{A}\mathbf{B}]_{ji} = \sum_k A_{jk} B_{ki}$
- $[\mathbf{B}^T\mathbf{A}^T]_{ij} = \sum_k [\mathbf{B}^T]_{ik} [\mathbf{A}^T]_{kj} = \sum_k B_{ki} A_{jk}$
- These are the same sum, so the matrices are equal

4.3 Determinants as areas

- $\det(\mathbf{M}) = 2 \times 3 - 1 \times 1 = 5$
- The unit square is mapped to a parallelogram of area 5, with no reflection since the determinant is positive

4.4 Expectations

- $\mathbb{E}[X] = \frac{1}{6}(1 + 2 + \dots + 6) = 3.5$
- $\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = 91/6$
- The variance is $91/6 - 49/4 = 35/12 \approx 2.92$
- For a discrete variable $\mathbb{E}[aX + b] = \sum_x \mathbb{P}[X = x] (ax + b) = a \sum_x \mathbb{P}[X = x] x + b \sum_x \mathbb{P}[X = x] = a \mathbb{E}[X] + b$, using $\sum_x \mathbb{P}[X = x] = 1$ (the continuous case is identical with integrals)